FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2020

ANSWERS, HINTS & SOLUTIONS

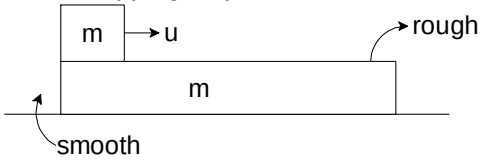
Physics

PART – I

SECTION – A

1. B

Sol. Where slipping stops both moves with same speed



$\therefore\ mu = 3mv \quad \therefore v = \frac{u}{3}.$

$\therefore$ W.D by friction $= \Delta KE$

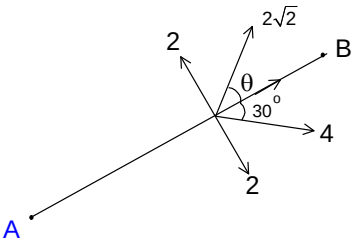
$$= \left[\frac{1}{2}2m\left(\frac{u}{3}\right)^2 + \frac{1}{2}m\left(\frac{u}{3}\right)^2\right] - \frac{1}{2}mu^2$$

$$= -\frac{1}{3}mu^2 \text{ Joule.}$$

2. C

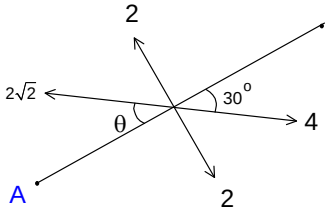
Sol. (i) $2\sqrt{2}\sin\theta = 2$

$\Rightarrow \theta = 45^o$



$$\Rightarrow t_1 = \frac{2\left(\sqrt{3}+1\right)}{4\frac{\sqrt{3}}{2}+2}$$

(ii) $2\sqrt{2}\sin\theta = 2$



$\theta = 45^o$

$$\Rightarrow t_2 = \frac{2\left(\sqrt{3}+1\right)}{4\frac{\sqrt{3}}{2}-2}$$

3. C

Sol. Time period of oscillation changes as it depends upon mass. Amplitude of the oscillation does not change, as another object is attached at extreme position. Total energy does not change as at extreme position, energy is in the form of potential energy. K.E. does not change, but maximum speed changes as mass changes.

4. B

Sol. By wien's law

$\lambda_m T = \text{constant}$

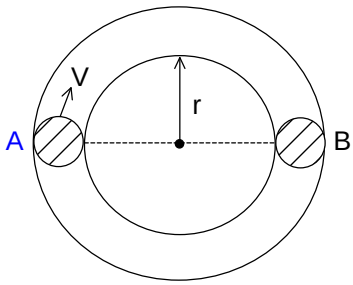
$\text{Power} \propto \left(T^4\right)$

5. C

Sol. $\because \pi r = VT$

$$e = \frac{V_B - V_A}{V-0}$$

$\Rightarrow V_B - V_B = eV$



one sphere will overtake the other after travelling a distance = $2\pi r$.

$$\therefore \Delta t = \frac{2\pi r}{eV} = \frac{2VT}{eV} = \frac{2T}{e}$$

6. D

Sol. Let g = 0 at x from M_1

$$\Rightarrow \frac{GM}{x^2} = \frac{GM_2}{(r-x)^2} \Rightarrow x = \frac{r}{3}$$

$\Rightarrow$ V at $x = \frac{r}{3}$ is

$$V = -\frac{GM_1}{x} - \frac{GM_2}{(r-x)}$$

$$\Rightarrow V = (-)\frac{9Gm}{r}$$

7. A

Sol. $\because I_{sphere} = I_{disc}$

$$\Rightarrow \frac{2}{5}MR^2 = M\frac{r^2}{2} + Mr^2$$

$$\Rightarrow \frac{r}{R} = \frac{2}{\sqrt{15}}$$

8. C

Sol. From conservation of mechanical energy

$$\frac{1}{2}Kx^2 = mg(h+x)$$

$$\Rightarrow K = 1000\,N/m$$

9. C

Sol. $\because \nu_{open} = \nu_{closed}$

$$\Rightarrow \frac{v}{2\ell_1} = \frac{v}{4\ell_2} \Rightarrow \frac{\ell_1}{\ell_1} = 2:1$$

10. C

Sol. For process $3 \to 1$

$P_1V_1 = P_3V_3 \Rightarrow V_3 = 4\ m^3$

For process $2 \to 3$

$P_2V_2^{\gamma} = P_3V_3^{\gamma} \Rightarrow V_2 = \sqrt{2}\,m^3$

For process $1 \to 2$

$W_{12} = P_1(V_2 - V_1) = \left(\sqrt{2}-1\right)4\times10^5\ J$

For the process $2 \to 3$

$$W_{23} = \frac{P_2V_2 - P_3V_3}{(\gamma - 1)} = 12\left(\sqrt{2}-1\right)\times10^5\ J$$

For the process $3 \to 1$

$$W_{31} = -P_1V_1\ln\left(\frac{V_3}{V_1}\right)$$

$$= -4\times10^5\times1\times\ln\left(\frac{4}{1}\right)$$

$\Delta U = 0$

For the process $1 \to 2 \to 3 \to 1$

$\Delta Q = \Delta U + \Delta W$

$= 0 + W_{12} + W_{23} + W_{31}$

$= 1.08\times10^5\ J$

11. A

Sol. $$\frac{\Delta v}{V} = \frac{\frac{M}{\rho'} - \frac{M}{\rho}}{\frac{M}{\rho}} = \frac{\rho - \rho'}{\rho'} \qquad .....(1)$$

Bulk nodulus $K = \dfrac{-P}{(\Delta V/v)}$ or $\dfrac{\Delta V}{V} = \dfrac{-P}{K}$ .....(2)

from (1) & (2) $\dfrac{\rho'}{\rho} = \left(1 - \dfrac{P}{K}\right)^{-1} = 1 + \dfrac{P}{K}$

12. A

Sol. Let ℓ is the length on potentiometer $i = \frac{5V}{5R} = 1A$

$(1.5\ell)A = 3 \Rightarrow$ † $= 2m$

13. D

Sol. $r = \sqrt{(8-2)^2 + (-5-3)^2} = 10$

$E = \frac{9\times10^9 \times 50\times10^{-6}}{100} = \frac{9}{2}\times10^3 = 4500\,V/m$

14. B

Sol. Place another identical hemisphere on given one, so that flat faces of two coincide. The total flux linked with cross- section of bottom half (lower hemisphere) is $\frac{Q}{2\in_0}$ as charge enclosed by given hemisphere is zero. So flux linked with flat face of hemisphere is $\frac{-Q}{2\in_0}$

15. C

Sol. $q_0 = 160\mu c$

$c = 6\mu F$

$q = 160e^{t8/3\times6}$

$q = 160\times e^{-t/16}$

$I = \frac{160}{16}e^{-t/16}$

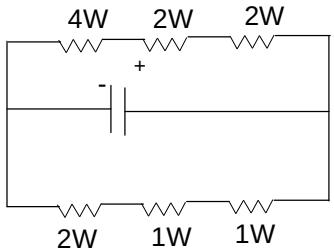
$I = 10e^{-t/16}$

$I(16) = 10e^{-1} = \frac{10}{e}$

$I_1 \times 4 = \left(\frac{10}{e} - I_1\right)\times 8$

$12I_1 = 80/e$

$I_1 = \frac{80}{12e} = \frac{20}{3e}A$



16. D

Sol. Magnetic field induction due to circular Arc $B_o = \frac{\mu_o I}{4\pi R}\cdot\phi$

Here $\phi_1 = \frac{\pi}{2}, \phi_2 = \frac{3\pi}{2}$

17. B

Sol. The current in wire AB, increase the flux through circular loop. Hence current in circular loop will be anticlockwise.

18. B

Sol. $emf = VBl$

$B = \frac{\mu_o I}{2\pi r}$

$$emf = V \cdot \frac{\mu_o I}{2\pi r} \cdot \ell = \frac{\mu_o V I \ell}{2\pi r}$$

19. A

Sol. Electric field and magnetite field will be parallel and velocity will also parallel to electric filed

$\Rightarrow$ All three are parallel

Hence only electric force will act on it

20. A

Sol. $r \propto n^2$

and, $L \propto n \quad \Rightarrow \omega r^2 \propto n$

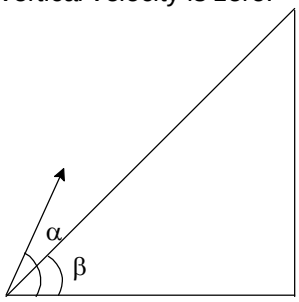
$\therefore T \propto \omega^{-1} \propto n^3$

So, $n_1 = 2n_2$

SECTION – B

21. 2

Sol. As the particle strikes the plane horizontally so its velocity is parallel to the horizontal axis and vertical velocity is zero.



$0 = u\sin\alpha - gt \qquad .....(i)$

$$u\sin\alpha = g\left[\frac{2u\sin(\alpha-\beta)}{g\cos\beta}\right]$$

$\Rightarrow \sin\alpha\cos\beta = 2\sin\alpha\cos\beta - 2\cos\alpha\sin\beta$

$\Rightarrow 2\tan B = \tan\alpha \Rightarrow \tan\beta = \frac{\tan\alpha}{2}$

22. 1

Sol. $W = \Delta K.E \qquad (1Kg).3m/s = (3kg).v$ by momentum conservation

$W = \frac{1}{2}(2)(1)^2 = 1$ Joule $\qquad v = 1 m/s$

SECTION – C

23. 00005.00

Sol. By phasor diagram

Amplitude of resulting waveform = $\sqrt{19}$A

X = 5

24. 00009.00

Sol. $\because \ \dfrac{1}{F}=\dfrac{1}{f_1}+\dfrac{1}{f_2}$

$$\Rightarrow \frac{1}{24}=\left[\left(\frac{\mu_1}{\frac{4}{3}}-1\right)\left(\frac{1}{20}+\frac{1}{20}\right)+\left(\frac{\mu_2}{\frac{4}{3}}-1\right)\left(\frac{1}{-20}-\frac{1}{20}\right)\right]$$

$$\Rightarrow (\mu_1-\mu_2)=\left(\frac{5}{9}\right).$$

25. 00016.00

Sol. $I_o=K(2A)^2$

Later, $A_1=0.6\,A,\ A_2=A$

So, $I_{max}=K(1.6\,A)^2=0.64I_o$

& $I_{min.}=K(0.4\,A)^2=0.04I_o$

$\therefore I_{max}:I_{min}=16$

Chemistry

PART – II

SECTION – A

26. C

Sol. Effective number of carbon atoms present at corners = $\frac{1}{8}\times 8 = 1$

Effective number of carbon atoms present at face centers = $\frac{1}{2}\times 6 = 3$

Effective number of carbon atoms present in tetrahedral holes = $8 \times \frac{1}{2} = 4$.

$\therefore$ Total number of effective carbon atoms present in a unit cell of diamond is **8**.

27. A

Sol. $Ti + 2I_2 \xrightarrow{500K}$ TiI_4 (Volatile stable compound) $\xrightarrow{1700K}$ Ti (Pure metal) $+ 2I_2$

28. D

Sol. $-OH \longrightarrow$ strongly activating group because of +R effect.
$-Cl \longrightarrow$ weakly deactivating group because of –I effect.
$-NO_2 \longrightarrow$ strongly deactivating group because of –R effect and –I effect.
So, 4 > 1 > 3 > 2.

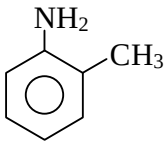
29. C

Sol. At anode: $\frac{1}{2}H_2(g) \longrightarrow H^+(aq.) + e^-$
At cathode: $AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq.)$
$\frac{1}{2}H_2(g) + AgCl(s) \longrightarrow H^+(aq) + Cl^-(aq) + Ag(s)$.

30. C

Sol. As_2S_3 sol is negatively charged and hence the ion with maximum positive charge shall be required in minimum concentration to cause its precipitation.

31. C

Sol.  is the weakest base as its conjugate acid is the least stable because of steric repulsion experienced by extra proton and existing $-CH_3$ group at ortho position.

32. A

Sol. $Fe(OH)_3 + NaOH \longrightarrow$ no reaction. Other hydroxides make soluble mettalate in cases of $Zn(OH)_2$ and $Al(OH)_3$ or mettalite in case of $Sn(OH)_2$.

33. D

Sol. Step growth polymers (or condensation polymers) are prepared by reaction between two functional groups and thereby eliminating small molecules during polymerization. Nylon-66 is an example of step growth polymer.

34. A

Sol. $K_c = \dfrac{[Z]}{[X][Y]}$

$[X] = \dfrac{1}{2}[Y] = \dfrac{1}{2}[Z] = a$ (let)

$\therefore$ [Z] = 2a, [Y] = 2a, [X] = a

$\therefore$ $10^4 = \dfrac{2a}{a \times 2a}$

$a = 10^{-4}$

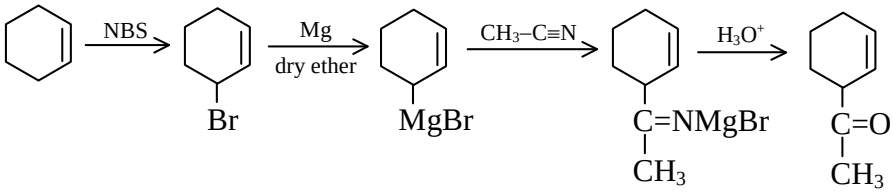
$\therefore$ $[Z] = 2a = 2 \times 10^{-4}$ mol L^{-1}.

35. C

Sol. $\Delta T_f = i \times K_f \times m = T_f^0 - T_f$

for urea and glucose, $i = 1$

36. B

Sol. 

37. A

Sol. Only O_2 has unpaired electrons in molecular orbital.

38. C

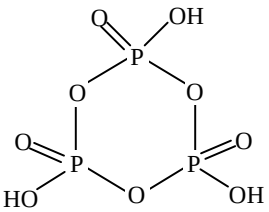
Sol. This is a Perkin reaction. Aromatic aldehyde which does not have α hydrogen reacts with alkanoic anhydrides in presence of alkanoate give $\alpha,\beta-$ unsaturated acid.

39. A

Sol. $pK_a = x < y < z$

Closer the –I group, stronger the acid.

40. C

Sol. 

There are three P–O–P bonds.]

41. C

Sol. In the given two parallel first order reactions

$\dfrac{[B]}{[C]} = \dfrac{k_1}{k_2} = \dfrac{2.50 \times 10^{-4}}{12.5 \times 10^{-5}}$

Percentage of C = $\frac{k_2 \times 100}{k_1 + k_2} = \frac{1.25 \times 10^{-4} \times 100}{3.75 \times 10^{-4}}$ = 33.3%.

42. A

Sol. $NO_3^- + 3Fe^{+2} + 4H^+ \longrightarrow NO + 3Fe^{+3} + 2H_2O$

$\left[Fe(H_2O)_6\right]^{+2} + NO \longrightarrow \underset{\text{Brown ring}}{\left[Fe(H_2O)_5(NO)\right]^{+2}} + H_2O$

43. D

Sol. Both have same molecular mass.

44. A

Sol. Compound 'A' give positive test with Na metal. That means it contains polar Hydrogen i.e., alcoholic group. It gives negative test with neutral $FeCl_3$ implies it does not contain of a phenolic group. It gives positive Lucas test means it is an alcohol.

45. A

Sol. $A = Ph - NH_2$

$B = NH_3$

SECTION – B

46. 4

Sol. Electronic configuration: $1s^2 2s^2 2p^6 3s^2 3p^3$

Number of occupied orbitals = 9

Number of occupied subshells = 5

47. 2

Sol. Two geometrical forms are possible. (cis + trans)

SECTION – C

48. 00069.16

Sol. Let the mass of acetaldehyde is *x* g, then mass of ethanol is (0.535 – *x*)g. When acetaldehyde is heated with Fehling solution, it gives 1.2 g of a red precipitate.

$CH_3CHO + 2CuO \rightarrow CH_3COO^- + \underset{\text{Red ppt.}}{Cu_2O}$

$\because$ 143 g of Cu_2O is coming from = 44 g of CH_3CHO

$\therefore$ 1.2 g of Cu_2O is coming from $\frac{44}{143} \times 1.2$ = 0.37 g of CH_3CHO

% of $CH_3CHO = \frac{0.37}{0.535} \times 100$ = 69.16 %

49. 00045.92

Sol. Photochlorination of 2,2–dimethylbutane gives three products as shown below

$$\underset{\text{(A)}}{CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3} + Cl_2 \xrightarrow{h\nu} \underset{\text{(B)}}{CH_3-\overset{\overset{CH_2Cl}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_2-CH_3} + \underset{\text{(C)}}{(CH_3)_3C-CHCl-CH_3} + (CH_3)_3C-CH_2-CH_2Cl$$

Reactivity factor × Probability factor (no. of H) = Number of parts

(A) :	1	×	9	=	9
(B) :	3.8	×	2	=	7.6
(C) :	1	×	3	=	3
	Total number of parts			=	19.6

The compound is the major product. Its percentage is given by

Percentage of (A) = $\frac{9\times100}{19.6}$ = 45.92**%**.

50. 00009.25

Sol. When half of the ammonium hydroxide is neutralised a base buffer is formed.

$$pOH = pK_b + \log\frac{[Salt]}{[Base]}$$

= 4.75 + log(1)

pH = **9.25**

Mathematics

PART – III

SECTION – A

51. C

Sol. $\because |z| = |z-1|$

$\therefore |z|^2 = |z-1|^2 \Rightarrow z\bar{z} = (z-1)(\bar{z}-1) \Rightarrow z+\bar{z} = 1$

Also, $|z| = |z+1|$

$\therefore |z|^2 = |z+1|^2 \Rightarrow z+\bar{z} = 1$

$\therefore |z+\bar{z}| = 1$

52. B

Sol. Let $y = 3-6x-8x^2$

$\therefore 8x^2+6x+(y-3) = 0$

for real x

$D \geq 0$

$\therefore y \leq \dfrac{33}{8}$

53. B

Sol. Required no. of ways = $4_{C_2} + 4_{P_2} + 32 = 50$

54. C

Sol. $\because 7^9 + 9^7 = (1+8)^7 - (1-8)^9$

55. A

Sol. Let the sum of n terms exceed 1000.

$\therefore \dfrac{1(1-3^n)}{1-3} > 1000 \Rightarrow 3^n > 2001$

$\therefore n = 7,8,9.....$

56. C

Sol. $\lim\limits_{n\to\infty}\left(\cos\dfrac{x}{n}\right)^n = e^{\lim\limits_{n\to\infty} n\left(\cos\frac{x}{n}-1\right)}$

$= e^{\lim\limits_{n\to\infty} -n.2\sin^2\left(\frac{x}{2n}\right)}$

$= e^{-2\cdot\lim\limits_{n\to\infty}\frac{x^2}{4n}} = e^0 = 1$

57. C

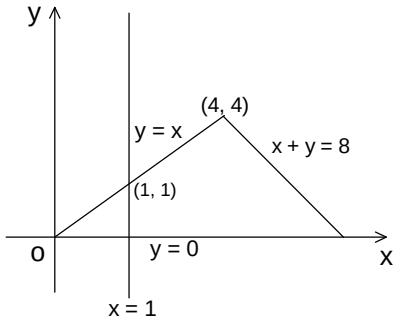
Sol. $\because x^3dx + \left(\frac{x}{y}\right)^2 \cdot \frac{ydx - xdy}{y^2} = 0$

$\therefore \int x^3dx + \int \left(\frac{x}{y}\right)^2 \cdot d\left(\frac{x}{y}\right) = \int 0$

$\therefore \frac{x^4}{4} + \frac{1}{3}\left(\frac{x}{y}\right)^3 = c'$

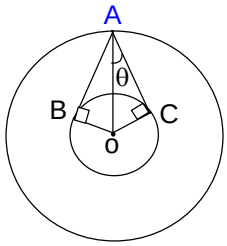
58. B

Sol.



59. C

Sol.



$\because \sin\theta = \frac{a\sin\alpha}{a} = \sin\alpha$

$\therefore \theta = \alpha$

$\therefore$ Required angle = 2θ = 2α

60. D

Sol. Do yourself

61. A

Sol. $u \times v = (a-b) \times (a+b) = 2a \times b$

$\therefore |u \times v| = 2|\vec{a} \times b| = 2\sqrt{\left[a^2b^2\sin^2\theta\right]} = 2\sqrt{\left(a^2b^2 - a^2b^2\cos^2\theta\right)} = 2\sqrt{\left[16 - (a-b)^2\right]}$

62. D

Sol. $Q.D. = \frac{40-20}{2} = 10$

63. C

Sol. $\because \sin x = 4\cos^3 x - 3\cos x$

$\therefore \sec^2 x.\tan x + 3\sec^2 x - 4 = 0$

$\therefore \tan^3 x + 3\tan^2 x + \tan x - 1 = 0$

$\therefore \tan x = -1, \quad \tan 2x = 1$

$\therefore x = \frac{3\pi}{4}, \frac{\pi}{8}, \frac{5\pi}{8}$

64. A

Sol. $\because$ secx is even function and $\log\left(x + \sqrt{1+x^2}\right)$ is odd function, therefore given function is even function.

65. A

Sol. $\Delta = \begin{vmatrix} 1 & 3\cos\theta & 1 \\ \sin\theta & 1 & 3\cos\theta \\ 1 & \sin\theta & 1 \end{vmatrix}$ *Apply* $R_3 \to R_3 \to R_1$

$= \left(3\cos\theta - \sin\theta\right)^2 \le 10$

66. A

Sol. $\because \lim_{x\to 0} \frac{f(x)-5}{x} = 4$

$= \lim_{x\to 0} \frac{f(x)-f(0)}{x} = 4$

$f'(0) = 4$

$\because g(x) = \left(x^2 + 2x + 3\right) f(x)$

$\therefore g'(x) = \left(x^2 + 2x + 3\right) f'(x) + (2x+2) f(x)$

$\therefore g'(0) = 3f'(0) + 2f(0) = 22$

67. C

Sol. $\because f(x) = 1$ *in* $0 < x < \frac{\pi}{2}$ (as [Cosx] = 0)

$\therefore$ f(x) is continuous in $\left[0, \frac{\pi}{2}\right]$

68. A

Sol. $\int \frac{dx}{\cos x\sqrt{\cos 2x}} = \int \frac{dx}{\cos x\sqrt{\frac{1-\tan^2 x}{1+\tan^2 x}}}$

Let tan x = t

$= \int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1}(\tan x) + c$

69. A

Sol. $\int_0^{2[x]} \left(x-[x]\right)dx = \int_0^{2(x).1} \left[x-(x)\right]dx = 2(x)dx\int_0^1 \left[x-(x)\right]$

$\because$ x – (x) is periodic function with period = 1

$= 2(x)\left[\frac{x^2}{2}\int_0^1 - \int_0^1 xdx\right] = 2(x)\left(\frac{1}{2}-0\right) = (x)$

70. C

Sol. P(5 or 6 or 7) in one draw = $\frac{3}{7}$

$\therefore$ Probability that in each of 3 draws, the unit bears 5 or 6 or 7 = $\left(\frac{3}{7}\right)^3$

SECTION – B

71. 1

Sol. $\left[\sqrt{1+x^2}-x\right]^{-1} = \frac{1}{\sqrt{1+x^2}-x}\times\frac{\sqrt{1+x^2}+x}{\sqrt{1+x^2}+x} = \left(1+x^2\right)^{\frac{1}{2}}+x$

$\therefore$ co – eff of x = 1

72. 2

Sol. $\because x-2 = 2^{\frac{2}{3}}+2^{\frac{1}{3}}$

$\therefore (x-2)^3 = 6+6(x-2)$

$\therefore x^3-6x^2+6x = 2$

SECTION – C

73. 00012.00

Sol. $F'(4) = \lim_{h\to 0}\frac{F(4+h)-F(4)}{h} = \lim_{h\to 0}\frac{F(4+h)-F(4+10)}{h}$

$= \lim_{h\to 0}\frac{2F(4).F(h)-2F(4).F(0)}{h}$

$= 2F(4)\lim_{h\to 0}\frac{F(h)-F(0)}{h} = 2F(4).F'(0)$

$= 2\times 2\times 3 = 12$

74. 00000.28

Sol. $\text{Required Prob.} = \frac{6C_2}{6C_1\times 9C_1} = \frac{15}{54}$

75. 01215.00

Sol. Let a = x – 1, b = y – 1, c = z – 1, d = w

then,

3a + b + c + d = 25 , where a, b, c, d $\geq$ 0----------(i)

Clearly $0 \leq a \leq 8$. If a = k, then

b + c + d = 25 – 3----------(ii)

No. of non-negative integral solution of equation (ii)

$$n+r-1_{C_r} = 3+25-3k-1_{C_{25-3k}}$$

$$= 27-3k_{C_2}$$

$$= \frac{3}{2}\left(3k^2-53k-234\right)$$

$\therefore$ Required number

$$= \frac{3}{2}\sum_{k=0}^{8}\left(3k^2-53k-234\right) = 1215$$